

The Science of Large Language Models

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Abstract. In the last few years there have been significant achievements and breakthroughs in large language models (LLMs). They have been integrated into various technologies and tools, such as search engines, chatbots, coding assistants, etc. These technologies are used by millions of users around the world. As a new key component of nowadays technology, it is important to study these gigantic computational models more deeply and from different aspects.

In this article, we promote the idea of defining a science for large language models. We highlight key aspects of large language models that should be studied in this specific branch of science. Some important dimensions include the limitations and shortcomings of LLMs, benchmarking and evaluation of LLMs, scaling laws in LLMs, physics of language models, biology of language models, physiology of language models, mathematics of language models, societal impacts of LLM-based AI technologies, etc.

Keywords: Science of Large Language Models, Science of LLMs, Generative AI, Scaling Laws, AI Agents, Agentic AI.

1 Introduction

In the last decade, there have been significant and unprecedented achievements in language modeling as well as in artificial intelligence. LLMs have sparked a new wave of innovation in natural language processing (NLP). In some ways, they have opened a new era in artificial intelligence, e.g., generative AI and agentic AI. This new wave of innovation in language models was inspired by the introduction of the Transformer architecture in 2017. After the Transformer architecture, there have been a series of milestones, particularly GPT series, that culminated in the advent of ChatGPT in late 2022. Since then leading AI companies have been developing various language models [1]. There have also been startups which have come up with novel developments in LLMs, e.g., DeepSeek-AI with a series of cost-efficient language models, i.e., DeepSeek-r1.

Large language models are now so advanced that they can solve mathematical problems at Olympiad levels, e.g., International Mathematical Olympiad (IMO) [2]. LLMs have also empowered various tools, such as search engines, chatbots, and coding assistants for programmers, etc. The breakthroughs in large language models have caused a leap in artificial intelligence and LLM-based AI technologies. Modern AI technologies are revolutionizing various fields such as software development, automation, healthcare, etc. It is important to understand the real impacts of AI in various domains. Besides their strengths,

LLMs have also their own shortcomings and limitations. They can easily hallucinate and generate factually wrong outputs. Furthermore, it turns out that nondeterminism is part of their nature [3].

Due to the widespread applications of LLMs, it is important to study these models thoroughly and more deeply. Motivated by the popularity of LLMs, in this article we promote the idea of defining a science for language models¹ [4]. We discuss some of key aspects of large language models that can be studied in this specific branch of science. The science of large language models help us better understand the gigantic mathematical models that are expected to have significant impacts in nowadays technological world. Furthermore, this science will hopefully enable us to enhance some critical dimensions of LLMs including AI safety and value alignment.

2 Some Keys Aspects of LLMs

In this section we highlight some key aspects of LLMs that can be studied in the science of large language models.

2.1 Limitations of LLMs

Despite the strengths of large language models in language-related tasks, LLMs seem to have some fundamental limitations. One of the notable shortcomings of LLMs is that they may simply hallucinate and generate factually wrong outputs (see e.g., Section 4.1 in [1]). Another limitation of LLMs is that they require large training datasets and consume significant compute resources. In [1], we have highlighted the shortcomings of LLMs. Besides these, large language models are based on complex network architectures and deep neural networks. They typically have billions of parameters, which makes it difficult to mathematically interpret how such gigantic models work. Researchers have been trying to develop ideas and frameworks that can help us better understand the internal workings of LLMs. In the following sub-sections we discuss some of these key ideas.

2.2 The Physics of Large Language Models

Large language models are gigantic computational models that can be studied from different perspectives. As effective tools for processing text and language, it is interesting and important to understand how these models process data, store and retrieve knowledge, and how they generate their outputs. Researchers in [5] and [6] have studied the physics of language models. They have analyzed the internal workflow of language models and how LLMs reason. The physics of LLMs is important as it tries to discover the universal laws of LLMs. Particularly, the physics of LLMs suggest dividing the concept of intelligence into several

¹ <https://aaai.org/conference/aaai/aaai-25/tutorial-and-lab-list/#TQ15> or <https://glaciohound.github.io/Science-of-LLMs-Tutorial/>

dimensions, including structures, knowledge, and reasoning [5]. These dimensions can be studied more deeply using different mathematical tools. Studying the physics of LLMs can help us better understand these models and potentially improve their shortcomings. It also allows us to have a better understanding of what LLMs can and cannot do [7] (see also [8]).

2.3 The Biology of Large Language Models

Motivated by the physics of language models, researchers in [9] have discussed the biology of large language models. They have investigated various phenomena in LLMs, including reasoning, planning, chain-of-thought (CoT), etc. Analogizing LLMs to biological structures and studying them from biological perspectives can be helpful for having deeper understanding of internal workings of LLMs. An interesting research question is to study and trace the thinking process in larger language models. For example, researchers in [9] studied the working mechanism of Anthropic’s Claude 3.5 Haiku using attribution graphs. The study shows that the model utilizes complex mechanisms, e.g., forward and backward planning, for generating its outputs [9].

2.4 The Physiology of Large Language Models

Apparently researchers have also highlighted studying LLMs even from physiological point of view [7]. In physiology of LLMs, one can study the role and functionality of different components in large language models.

Current large language models are typically based on the Transformer architecture. The Transformer architecture has been built on the attention mechanism, word embedding, positional encoding, etc. Therefore, to better understand the internal workings of LLMs and why they perform well, one can delve into the aforementioned sub-modules in the Transformer architecture. Particularly, embedding techniques have played a significant role in LLMs. Embedding techniques have been utilized at different levels. Besides word embedding, one can explore sentence embedding [10], concept embedding [11], rule embedding [12], etc.

2.5 The Mathematics of Large Language Models

Large language models are gigantic mathematical models with billions of parameters. Since they are already mathematical models, studying LLMs from mathematical perspective may seem pointless. However, there are various ways to study LLMs through mathematical and computational lenses, e.g., using sparse dictionary learning² [13]. One interesting line of research is to study how LLMs structure and store knowledge in hyper-dimensional vector spaces. Another research problem is to analyze the computational and attribution graphs in LLMs³.

² <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>

³ <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>

Studying LLMs from a mathematical perspective can be useful for different applications, e.g., mechanistic interpretability⁴. Mechanistic interpretability is an emerging research approach in explainable AI. It aims at understanding internal working of neural networks by reverse engineering the computations in neural networks⁵. It should be noted that interested readers may refer to the NeurIPS workshops on mechanistic interpretability⁶.

2.6 The Computation of Large Language Models

Large language models require significant amount of energy and compute resources for their training. Therefore, it is important to study large language models and LLM-based AI technologies from the computation point of view. In the computation of large language models, researchers can study how to optimize LLMs by reducing their energy consumption as well as their inference time. Particularly, matrix multiplication is an essential part of neural networks as well as the Transformer architecture. Designing efficient matrix multiplication methods can significantly improve LLMs (see e.g., [14]). Another interesting line of research is to study how LLMs may result in the raise of novel post-transformer and new compute architectures (see e.g., Section 4.7 in [1]).

2.7 Evaluation of Large Language Models

Leading AI companies have developed various large language models and chatbots. AI technologies, particularly generative and agentic AI based on LLMs, are anticipated to have significant societal impacts. It is important to assess such technologies and ensure they align with human values. In addition, AI technologies should follow standard AI principles/guidelines [15]. Researchers have designed various benchmarking and evaluation methodologies for evaluating LLMs (see, e.g., Table 9 in [1] and references therein). As companies are competing on developing more advanced AI technologies, researchers and engineers should focus on evaluating state-of-art large language models and AI technologies. For more details on various benchmarks for LLMs evaluation, interested readers may refer to Section 4.4 of [1].

2.8 Scaling Laws in Large Language Models

Scaling laws, e.g., power law distribution and Chinchilla scaling, are essential in training large language models. They are useful for describing the relation between the performance of a model and other contributing factors, i.e., the model's size, the size of training data, and the amount of compute resources. Until recent years and for the most part, there has been some simple scaling laws

⁴ <https://www.transformer-circuits.pub/2022/mech-interp-essay>

⁵ <https://transformer-circuits.pub/2021/framework/index.html>

⁶ <https://neurips.cc/virtual/2025/workshop/109547> and <https://icml.cc/virtual/2024/workshop/29953>

in the field of artificial intelligence. That is, increasing the number of parameters of a model along with training data and computational resources improves the model’s capability. However, recent advancements in large language models have shown that other scaling laws may emerge in the world of large language models. Some notable scaling laws include pre-training scaling laws, post-training scaling laws, and test-time compute scaling.

Among the aforementioned scaling laws, the post-training and test-time scaling laws can be helpful for improving large language models or even developing more efficient models. Particularly, test-time scaling laws can be helpful for improving the reasoning capabilities of language models. In fact, these scaling laws along with other techniques such as model or knowledge distillation can be used for transferring knowledge from very large LLMs to distilled small language models (SLMs).

There are other important scaling laws in large language models, e.g., knowledge capacity scaling laws [6] and asymmetry of verification and verifier’s law [16]. The knowledge capacity scaling laws can give us thresholds or bounds on the amount of knowledge which can be stored in each parameter of the model. In [16], researchers claimed that language models can store 2 bits of knowledge per parameter. The verifier’s laws relates the solvability of tasks using AI with their verifiability [16]. This principle states that all tasks which are possible to solve and easy to verify will eventually be solved by artificial intelligence [16].

2.9 The Engineering of Large Language Models

Most of the recently developed large language models are based on the Transformer architecture. Transformer-based LLMs have demonstrated extraordinary abilities in language related tasks. However, this architecture has its own strengths and weaknesses. There has been ongoing research for designing alternative architectures that can lead to more efficient large language models. Particularly, state space models (SSMs) and RNN-based architectures have recently attracted researchers attention [1]. It is important to explore various alternatives for developing yet more efficient large language models. This can lead to saving significant amount of compute and energy resources. SSM-based and RNN-based architectures can also be useful for extending the content window of LLMs [1].

3 Language Models Science: The Science of Language Models

There are various fields, such as linguistics, natural language processing (NLP), and natural language understanding (NLU), that study language from different perspectives. However, large language models (including LLM-based AI technologies) seem to be a new generation of language models arguably with emergent abilities [17]. Some experts claim that generative AI technologies can have significant societal impacts, e.g., using agentic AI in work automation. It is

necessary to study large language models as stand-alone complex systems and gigantic computational models from different angles.

Particularly, it is interesting to understand how LLMs store and retrieve knowledge, how they reason and plan, and if they have any understanding of language. These can be studied in the so-called physics, biology and physiology of large language models. Researchers have used mathematical methods to analyze the trace of thoughts and computational graphs in LLMs [9]. Scaling laws are another important aspect of large language models, that are important for designing more efficient LLMs.

The mathematics and computation of large language models can help us design more interpretable LLMs. Such models are essential for explainable AI (XAI). Interpretability and explainability of AI models are fundamental to AI safety. AI safety is an important subject in artificial intelligence. Considering the recent advancements in AI, it is expected that AI safety to become even more important. Yet, the computation of LLMs can provide us with insights on how LLMs can be improved from computational perspective, e.g., reducing computational costs and inference time in generative AI tools. Training LLMs is significantly costly. Achieving even minor improvements may result in significant savings in compute and energy resources.

It is also critical to study the societal impacts of LLM-based technologies. Some common perception about generative AI technologies are that they can significantly enhance productivity or they might replace some professions, e.g., software engineering. However, it turns out that the reality may be different⁷, see e.g., [18]. Besides the aforementioned aspects, there are other important dimensions of AI that are worth studying, e.g., AI in education, AI for healthcare, AI in finance and trading, etc. We expect to write more about the applications of AI in various domains in our future works. The aforementioned key sub-domains of large language models are summarized in Figure 1.

4 The Three Key Triangles in AI and LLMs

Current large language models and AI models need to be improved in several aspects. We categorize these aspects into three groups and call them the Three key Triangles in AI and LLMs. At a high level, these triangles cover key requirements, critical features or capabilities, and desired properties. Figure 2 shows the three key triangles in AI and LLMs. In what follows we discuss the importance of these dimensions in modern AI and LLMs.

The key requirements triangle highlights three minimum expectations that large language models and AI technologies should meet. While advancements in LLMs have largely moved from pre-training to post-training, efficient training

⁷ <https://bootcamps.cs.cmu.edu/blog/will-ai-replace-software-engineers-reality-check>

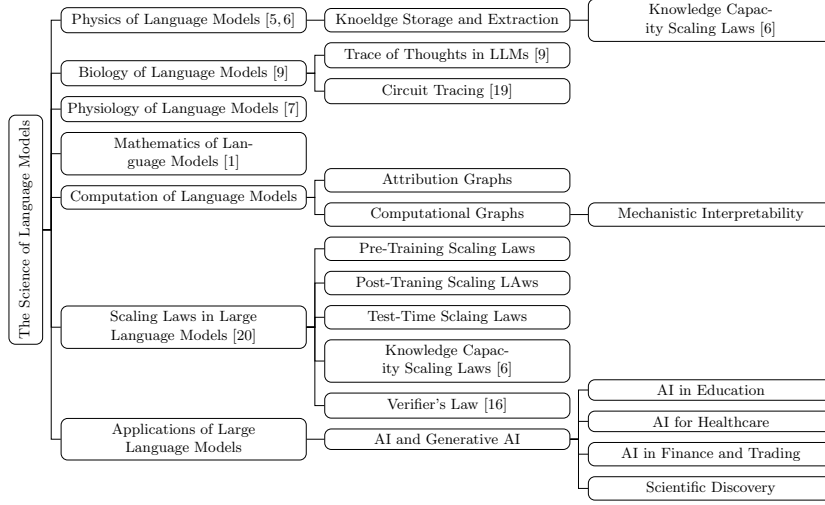


Fig. 1: The Science of Language Models

mechanisms are still a necessity for developing LLMs. More importantly, LLMs are expected to function at high performance but with low inference costs.

The critical features are essential functionalities that are needed for AI system to eventually reach to human-level intelligence. These features are still more or less missing in current LLMs. Particularly, AI technologies are still not that good in reasoning and are not fully reliable in planning. They also fall short of having consistent memory that can function similar to the memory in a human brain. In fact, the critical features are interesting open problems in AI and LLMs, that require further research. For example, memorization in LLMs and neural networks is an interesting research direction. This line of research is related to how much these models can remember and reveal from their training data [21]. It is important to design AI systems and LLMs that respect data privacy standards and guidelines.

Lastly, the desired properties triangle specifies some key characteristics that AI systems and LLMs should have. These properties include accuracy, trust & reliability, and security & privacy. In particular, accuracy is one of the key features in which current AI systems require significant improvement. According to the State of AI report 2025, by McKinsey, inaccuracy is the number one AI-related risk that users and organizations are trying to mitigate⁸. Besides accuracy, AI systems should be trustworthy and reliable. Furthermore, AI systems and LLMs are expected to provide mechanisms for ensuring data security and protecting users' privacy.

⁸ <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>

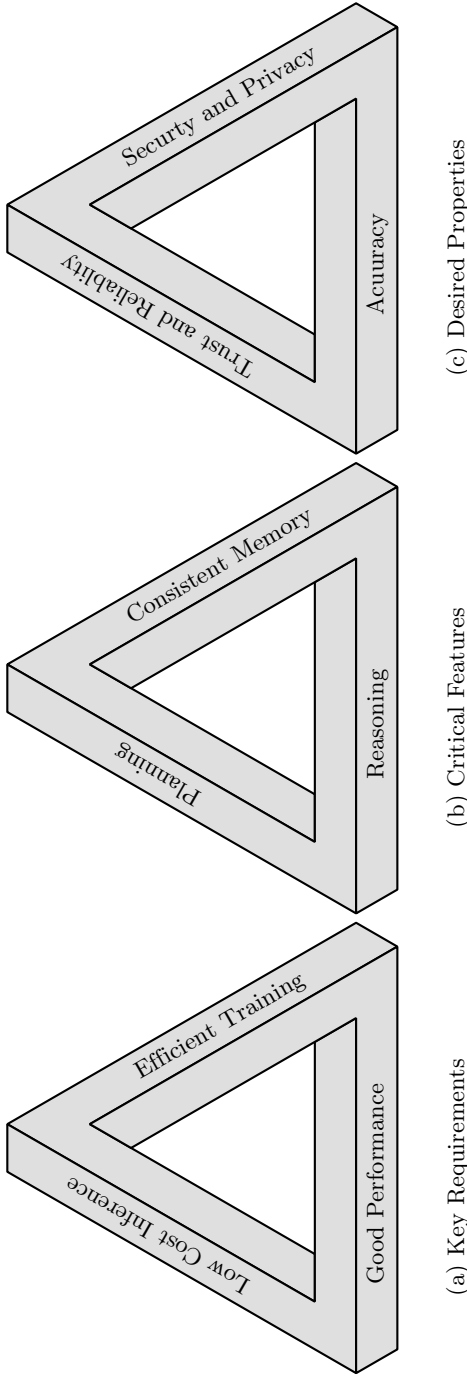


Fig. 2: The Three Key Triangles in AI and LLMs

5 Conclusion

In this article we promoted the idea of defining a science for large language models. The science of language models is for studying the gigantic computational models behind large language models more deeply and thoroughly. This is necessary for understanding the limitations of LLMs, addressing their shortcomings, and improving state-of-art large language models.

Large language models and generative AI technologies are expected to have significant societal impacts. We hope this article motivates researchers to study various aspects of LLMs in a scientific and systematic way. The scaling laws in LLMs and understanding the internal workings of LLMs are essential for developing explainable AI models. This in turn can lead to AI technologies that align universal human values and are beneficial for all.

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